

Some useful definitions

Events clustering: since multiple pipelines can trigger, more than one event can be associated with the same astrophysical event —> events that have a temporal proximity of $[-1,+1]$ from each other are clustered in a single **SUPEREVENT**

On GraceDB events are labeled with a G*** name, superevents with an S*** name

P_astro: Quantifies the probability that the event is of astrophysical origin

- Provided only for CBC pipelines
- Depend on our prior knowledge on the astrophysical rate of each class (BBH, NSBH, BNS) as well as on assumptions on the mass distribution
- Divided in $P(\text{BBH})$, $P(\text{NSBH})$, $P(\text{BNS})$, $P(\text{Terrestrial})$ and they sum to 1

Some useful definitions

EM_bright classification:

- P(HasNS) $\rightarrow$ at least one of the compact objects in the binary has a mass that is consistent with a neutron star $\rightarrow 1 M_{\text{sun}} < M < \mathbf{M_{\text{max}}}$ (EoS)
- P(HasRemnant) $\rightarrow$ Non-zero mass is left around the final compact object. Depends on the mass ratio, priors on spins, EoS $\rightarrow$ each pipeline can give different values
- P(HasMassGap) $\rightarrow$ one object in the 3-5 M_{sun} range

Coarse-grained Mass Information (CGMI):

- An array of chirp mass bins from $[0.1, \dots, 1000] M_{\text{sun}}$ with associated binned probability $[P(n, n+1)]$

$$\mathcal{M}_{\text{chirp}} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$